



Subject Vision and Intent: Design Technology

Our Vision

At King's Cross Academy, we believe that Design & Technology gives children the knowledge, skills and confidence to identify problems, develop ideas and create purposeful solutions. Our ambition is for children to leave KCA seeing themselves as designers: curious about how and why things work, willing to experiment and able to use their knowledge creatively to respond to real needs.

Design & Technology provides children with opportunities to combine creativity with practical thinking. Children learn that successful design begins with a purpose and a user. They consider needs, explore possibilities, make informed choices and learn through the process of creating, testing and improving their ideas.

Our Design & Technology curriculum reflects our Academy vision, *Love Learning Together*. Designing and making require children to demonstrate many of our KCA values: the courage to try something new, the endurance to persevere when an idea does not work, the aspiration to improve and the integrity to evaluate their own work thoughtfully.

Our Intent

Our Design & Technology curriculum is ambitious, inclusive and carefully sequenced. It identifies the essential knowledge, vocabulary, technical understanding and practical skills we want children to develop from the Early Years to Year 6, enabling them to become increasingly confident and independent designers.

Our curriculum is built around four interconnected elements: Designing, Making, Technical Knowledge and Evaluating.

Through Designing, children learn to identify a purpose and consider the needs of a particular user. They research existing products, generate and communicate ideas and develop increasingly detailed design criteria to guide their thinking.

Through Making, children learn to select and use appropriate tools, materials, components and techniques with increasing accuracy and independence. Practical skills are deliberately taught and developed over time so that children can draw upon an increasingly broad repertoire when bringing their designs to life.

Through Technical Knowledge, children develop an understanding of how products work. Their knowledge of structures, mechanisms, electrical systems, textiles, materials and other technical concepts becomes increasingly sophisticated as they progress through the school, allowing them to make informed decisions about their own designs.

Through Evaluating, children learn that designing is an iterative process. They test their ideas and products against their original purpose and design criteria, identify strengths and areas for improvement and use what they discover to refine their thinking. Children also investigate and evaluate existing products and the work of designers, inventors and engineers, considering how design responds to different needs and contexts.

Our curriculum deliberately moves beyond simply following instructions to produce a predetermined outcome. As children develop their knowledge and skills, they are expected to make increasingly independent choices, explain their decisions, solve problems and adapt their designs when things do not work as expected.

We want children to understand that there is rarely one perfect solution to a design problem. At KCA, the thinking, experimentation, problem-solving and evaluation that take place throughout the design process are as important as the finished product.

Every Child

We have high expectations for every child. Children with SEND, children who are disadvantaged and children who are developing English as an additional language access the same ambitious Design & Technology curriculum.

Teachers identify the essential knowledge and practical skills that all children need and adapt teaching so that barriers are reduced without lowering expectations. Explicit modelling, carefully sequenced practical instruction, visual resources, appropriate tools and materials, vocabulary support and opportunities for repeated practice enable children to participate successfully and develop increasing independence.

Design & Technology provides particular opportunities for children to experience success through practical learning, creativity and problem-solving. We ensure that every child has opportunities to handle materials, use tools, explore mechanisms and technologies, make decisions and experience the satisfaction of turning an idea into something purposeful.

By the end of Year 6, our aim is not simply that children can make successful products, but that they understand the process through which successful products are designed. We want them to identify needs, generate and communicate ideas, draw upon secure technical and practical knowledge, make purposeful choices and critically evaluate what they create.

Above all, we want children to leave KCA as resilient and resourceful designers who are unafraid of getting things wrong, understand that ideas improve through testing and refinement, and have the confidence to use their creativity and knowledge to solve problems.